FISH Testing Metafer Usage Windows and Optimization Recommendations

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Background:

Metafers are utilized within the FISH and Conventional Chromosomes labs to scan and image hematologic dropped cell suspension tests. For the FISH lab they image DAPI stained nuclei as well as red, green and aqua probes. For Conventional Chromosomes they are used to image metaphases for karyotyping. Each metafer frame can hold 5 slides (each containing two probed wells or sites). It takes approximately 5 minutes for each site to scan. The objective of this project is to define active scanning windows for the FISH lab. This would allow us to better define when Conventional Chromosomes can have access to our underutilized scanners, with the desired outcome of this project being optimized metafer utilization.

Methods:

Case scanning data was pulled from NEON, excluding all non-FISH test types, from 01/01/2026 to 07/20/2026. This produced 6571 rows of data following data cleaning, with the relevant features being case number, metafer number, capture date, capture time, and number of scans by case. The time feature was converted to binned 30-minute increments. The data was grouped by month to determine if there was a trend in daily total scans. The result was a clear weekly pattern that persisted month after month.

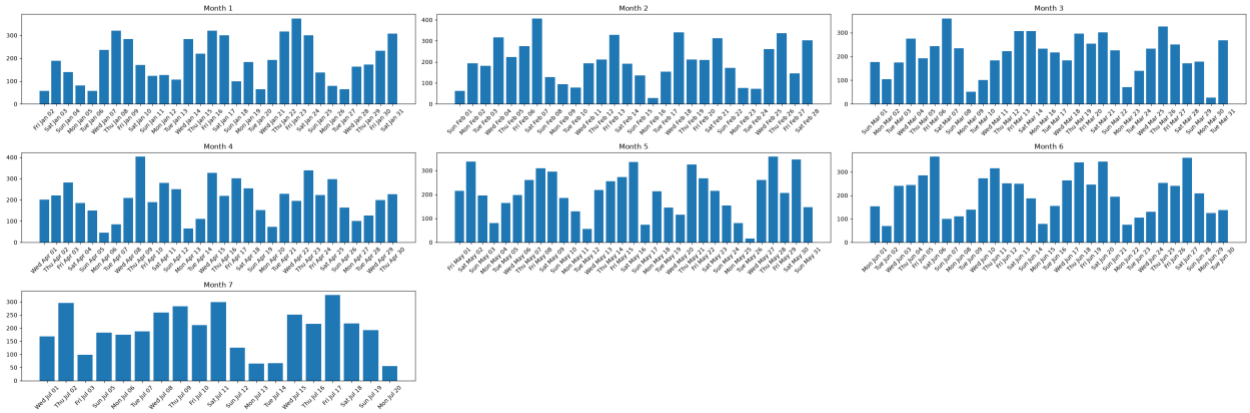


Figure 1**: Monthly distribution of completed scans across time intervals.** Bar plots show the number of completed scans recorded during each 30-minute time interval for Months 1–7 (January 1 – July 20). Each panel represents an individual month, with bars indicating scan volume within each time window.

The data was then grouped by weekday index and metafer to see if specific metafers were being utilized more than others. This was then further broken down by shift, to identify any shift dependent preferences in metafer utilization. The results indicated a slight but insignificant preference for meta 2, as well as an expected difference in scanner utilization on the Sunday AM shift due to a lack of weekend rotation during that specific day of the week. It was determined from this information that it was not necessary to include the shift or specific metafer each case was scanned on as features for training our unsupervised model.

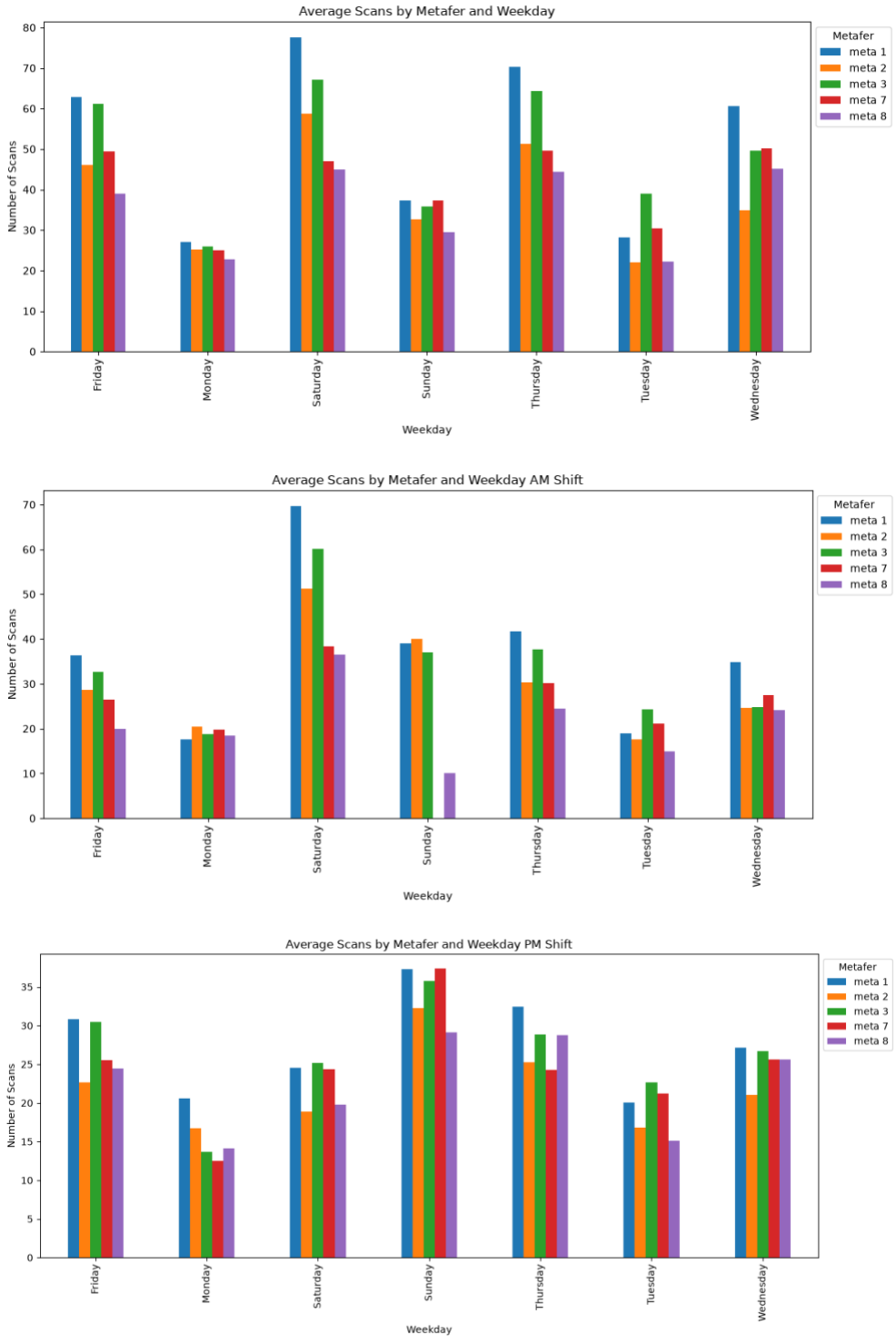
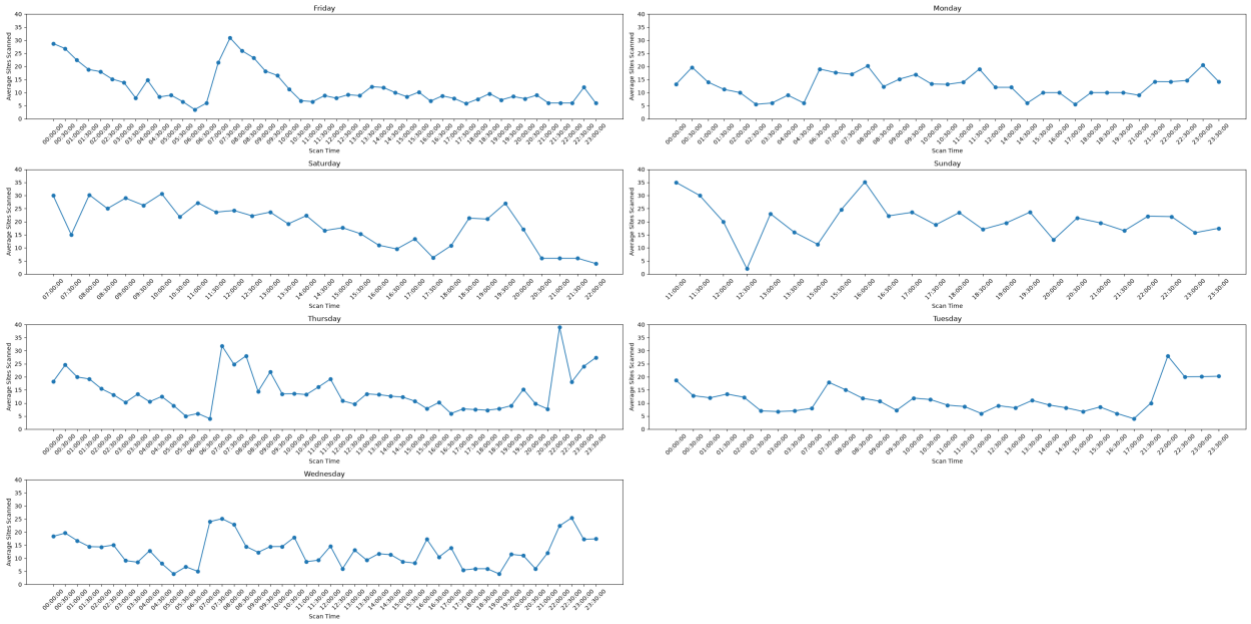


Figure 2: **Average scan volume by weekday, metafer, and shift.** Bar plots show the mean number of completed scans grouped by weekday index and stratified by metafer scanner. The top panel represents all scanning activity combined across shifts, while the middle and bottom panels represent average scan volume during AM and PM shifts, respectively. Each colored bar corresponds to an individual metafer scanner (meta 1, meta 2, meta 3, meta 7, and meta 8). The figure demonstrates variation in scanner utilization across days of the week and between instruments.

The data was then grouped by weekday and binned time intervals to attempt to visualize any specific trends. It appeared that there were specific active time windows that stood out for some days of the week, while for nearly half of the days of the week the trend was more difficult to visually interpret. This was partially aided by adding an active scanner threshold line at 12 scans. This threshold approximates the first quartile and indicates approximately the number of scans per 30-minute interval that would require more than 2 metafers to accomplish.

a.



b.

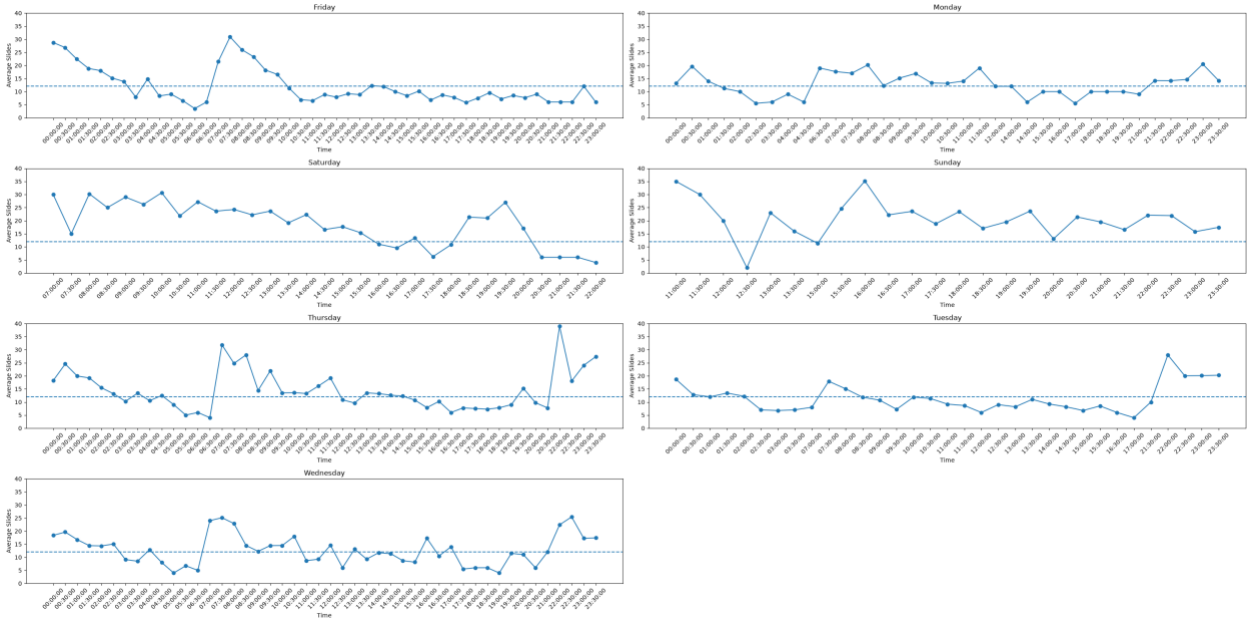


Figure 3: **Weekday distribution of average sites scanned across rolling 30-minute time bins.** Bar plots show the average scans completed during each 30-minut interval in a 24-hour period. Each panel represents a day of the week. Figure 3a includes a 12-site activity threshold.

An active scanner threshold feature was added to the data, which made it possible to establish preliminary active scanning windows. These windows could then be compared to the windows established by the trained DBScan model later in analysis.

weekday	window_id	start_time	end_time	avg_sites
Friday	1	00:00:00	03:00:00	20.55
Friday	3	04:00:00	04:00:00	14.75
Friday	5	07:00:00	09:30:00	22.75
Friday	7	13:30:00	13:30:00	12.27
Monday	9	00:00:00	01:00:00	15.63
Monday	11	06:30:00	11:30:00	16.14
Monday	13	21:30:00	23:30:00	15.53
Saturday	13	07:00:00	15:30:00	23.34
Saturday	15	17:00:00	17:00:00	13.38
Saturday	17	18:30:00	20:00:00	21.58
Sunday	19	11:00:00	12:00:00	28.33
Sunday	21	13:00:00	13:30:00	19.5
Sunday	23	15:30:00	23:30:00	20.93
Thursday	23	00:00:00	02:30:00	18.42
Thursday	25	03:30:00	03:30:00	13.4
Thursday	27	04:30:00	04:30:00	12.5
Thursday	29	07:00:00	11:30:00	19.66
Thursday	31	13:00:00	14:30:00	12.9
Thursday	33	19:30:00	19:30:00	15.25
Thursday	35	22:00:00	23:30:00	27.08
Tuesday	35	00:00:00	00:30:00	15.74
Tuesday	37	01:30:00	02:00:00	12.82
Tuesday	39	07:30:00	08:00:00	16.48
Tuesday	41	22:00:00	23:30:00	22.08
Wednesday	41	00:00:00	02:30:00	16.43
Wednesday	43	04:00:00	04:00:00	12.9
Wednesday	45	07:00:00	10:30:00	18.22
Wednesday	47	12:00:00	12:00:00	14.6
Wednesday	49	13:00:00	13:00:00	13.12
Wednesday	51	16:00:00	16:00:00	17.25
Wednesday	53	17:00:00	17:00:00	14.0
Wednesday	55	22:00:00	23:30:00	20.63

Table 1: **Average sites captured across preliminary time windows by weekday.** Table summarizing preliminary time window specifications (window_id) categorized by day of the week, window start time, and window end time. The table details the average number of sites captured for each defined temporal window prior to model training.

Following exploratory data analysis, the cleaned data was ingested by the model pipeline. StandardScalar was used to encode the numerical features, OneHotEncoder was used to encode the categorical features, and the time bins were encoded with a custom encoder (TimeToOperationalMinutes) that accounted for the PM shift crossing the midnight hour. The custom encoded time bins were then scaled using Standard Scalar. Hyperparameter tuning of the epsilon and minimum sample (up to 2% of the dataset) paramters for DBScan in the sklearn cluster package was then performed. Silhouette scores adjusted for noise fraction were then compared across all models and the best performance was chosen. The best model was determined by a silhouette score of 0.598 and noise fraction of 0.255. The clusters were then appended to the dataset, and new active windows were determined by grouping by cluster, day of the week and time windows.

cluster	weekday	start_time	end_time	avg_scans	observations
0	Friday	04:30:00	19:30:00	4.77	466
1	Saturday	07:00:00	20:00:00	4.63	1064
2	Saturday	08:00:00	15:00:00	12.19	206
3	Monday	04:30:00	12:00:00	5.57	176
4	Tuesday	07:00:00	13:00:00	5.76	122
5	Wednesday	00:00:00	23:30:00	5.58	308
6	Friday	00:00:00	23:00:00	5.32	346
7	Wednesday	05:30:00	18:30:00	4.44	355
8	Thursday	00:00:00	23:30:00	5.36	371
9	Thursday	04:30:00	19:00:00	4.68	502
10	Monday	00:00:00	23:30:00	5.78	130
11	Sunday	15:00:00	23:30:00	5.45	390
12	Tuesday	00:00:00	23:30:00	5.67	191
13	Friday	07:00:00	10:00:00	13.13	87
14	Thursday	07:00:00	11:30:00	13.28	95
15	Tuesday	00:00:00	23:30:00	1.46	84

Table 2: **Average scan volume and total observations by cluster, weekday, and time interval**. Table summarizing operational clusters (0 through 15) categorized by day of the week, window start time, and window end time. Columns report the average number of completed scans alongside total observation counts for each identified cluster.

Conclusion:

After interpreting clusters in conjunction with figure 3b, several active scanning time windows were established. During these active time windows, the average scans per 30-minute increment indicate a need for 2 to 3 scanners to accommodate caseload needs. During all other time windows, a single scanner should be sufficient to efficiently scan all FISH slides while mutually optimizing scanned caseload for the Conventional Chromosomes lab. The recommended active time windows will appear in table 3. The average amount of time it would take to scan the total caseload by shift and weekday was then incremented by the number of scanners used. The difference in the total time required to scan caseload was negligible, in the context of FISH lab operations, between 2-3 metafers being utilized and all 5 metafers being utilized(Table 4). In summation, these windows should serve as a baseline suggestion, while surges in incoming caseload should be considered daily for metafer utilization planning between FISH and Conventional Chromosomes. These windows can be re-evaluated a month after implementation and as necessary in the future.

Recommended Scanner Utilization Windows

Weekday	Start Time	End Time
Sunday	00:00	23:30
Monday	06:30	11:30
Tuesday	21:30	00:30
Wednesday	20:00	12:00
Thursday	20:00	12:00
Friday	00:00	10:00
Saturday	00:00	21:30

Table 3: **Recommended scanner utilization windows by weekday.** Table detailing optimal operational windows, specifying recommended Start Time and End Time for scanner usage on each day of the week (Sunday through Saturday). The schedule indicates targeted operational hours across most days—ranging from early morning shifts to late evening windows—with Tuesday noted as not applicable (N/A) as there is no indication of an active scanning window in the dataset.

a.

Average Day Shift Scans and Total Hours by Weekday

weekday	scans	total_hours_1_metafer	total_hours_2_metafers	total_hours_3_metafers	total_hours_4_metafers	total_hours_5_metafers
Friday	131.9	10.99	5.5	3.66	2.75	2.2
Monday	57.5	4.79	2.4	1.6	1.2	0.96
Saturday	245.61	20.47	10.23	6.82	5.12	4.09
Sunday	126.0	10.5	5.25	3.5	2.62	2.1
Thursday	150.89	12.57	6.29	4.19	3.14	2.51
Tuesday	61.19	5.1	2.55	1.7	1.27	1.02
Wednesday	113.75	9.48	4.74	3.16	2.37	1.9

b.

Average Night Shift Scans and Total Hours by Weekday

weekday	scans	total_hours_1_metafer	total_hours_2_metafers	total_hours_3_metafers	total_hours_4_metafers	total_hours_5_metafers
Friday	116.59	9.72	4.86	3.24	2.43	1.94
Monday	54.48	4.54	2.27	1.51	1.14	0.91
Saturday	62.44	5.2	2.6	1.73	1.3	1.04
Sunday	155.18	12.93	6.47	4.31	3.23	2.59
Thursday	120.64	10.05	5.03	3.35	2.51	2.01
Tuesday	75.71	6.31	3.15	2.1	1.58	1.26
Wednesday	113.68	9.47	4.74	3.16	2.37	1.89

Table 4: **Average time daily scans and time for scan completion by number of metafers used.** Table summarizing the amount of time in hours it would take to complete the average number of scans by day of the week, incremented by number of metafers used for day shift (a) and night shift (b).